

Pricing the DeFi Tail: Do Protocols or Depositors Price Operational Risk?

CBT 2026 — 10th International Workshop on

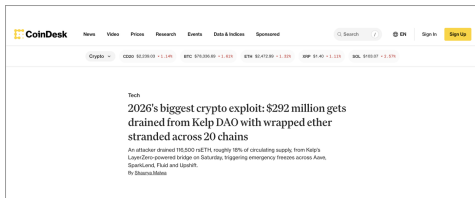
Cryptocurrencies and Blockchain Technology

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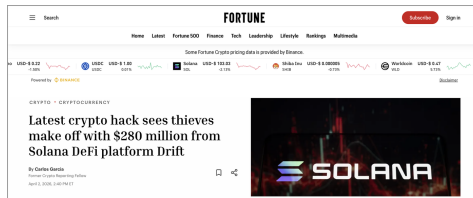
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The question

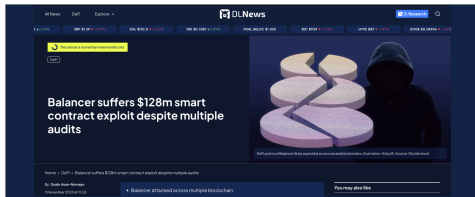
The losses are real, and they are recent



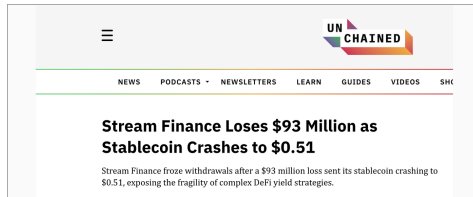
USD 292 m, Bridge — a forged cross-chain message



USD 290 m, Derivatives — admin approvals abused



USD 128 m, DEX — a rounding error, after audits



USD 93 m, Yield — no exploit at all

DeFi removes many risks — operational risk is not one of them

DeFi's architecture genuinely removes several bank-balance-sheet risks: smart-contract custody replaces the custodian, settlement is atomic, counterparty credit risk collapses into overcollateralization.

- **Operational risk remains** — losses from failed processes, people, systems and external events — and in some cases concentrates: contract code, admin keys, oracles, economic design.
- Unlike a bank, DeFi is **infrastructure**: no regulatory capital requirement, no deposit insurance, no resolution authority. Nothing stands between a loss and the depositor's claim unless a protocol chooses to put it there.
- Yet DeFi markets **earn products** — lending supply, yield vaults, stablecoin positions, crypto neobanks, etc.

The decisive difference

Not whether the risk exists — it does, and it is measurable — but *who bears it* and *whether anyone priced it*.

Two responses, neither disciplined by a regulator

The protocol can hold a buffer.

Aave's Umbrella safety module, the Sky surplus buffer, Compound's reserve factor — governance-owned reserves with the function Basel calls capital, sized by vote rather than by any loss model.

Or the depositor demands a premium.

A supply yield above the risk-free rate compensates the depositor for bearing the tail — the market-discipline mechanism by which uninsured bank creditors price institutional risk.

The two are **substitutes** for the same risk, chosen by different participants.

Research question

Is either response sized to the operational-risk tail it is meant to cover? Answering it needs a common benchmark — so I first have to quantify the tail.

Contributions

1. A **consolidated event dataset**: seven public feeds, deduplicated, filtered to depositor-facing DeFi losses, tagged by sector and Basel Level-1 event type — $n = 1,075$, USD 9.45 B, 2020–2026.
2. A **per-sector Basel LDA**, fitted as a bank supervisor would, and benchmarked against banking operational risk.
3. Both margins tested against that one benchmark: **protocol capital buffers** and **depositor risk premia**.
4. Evidence that the two **substitute** — in the right direction, at the wrong order of magnitude — and the policy that follows.

Headline

The four buffered top-10 Lending venues cover $\sim 5\%$ of their modeled $\text{VaR}_{99.9}$; unbuffered venues pay +54 bps against a tail worth 34–81% of TVL.

Data

Seven feeds, one deduplicated dataset

Source	Raw	In dataset
DefiLlama hack ledger	528	379
kismp123 DSI corpus	823	290
BlockSec Incidents Library	285	201
rekt.news leaderboard	281	219
DeFiHackLabs catalog	615	163
de.fi/rekt-database	4,030	704
SlowMist Hacked	2,100	639
Consolidated (dedup.)		1,164

- Median-across-sources loss reconciliation.
- Two-pass dedup: name within ± 14 d, then date ± 7 d and loss within $\pm 10\%$.
- Out-of-scope removed (CEX, memecoins, NFTs, phishing): 153 events.
- A **depositor-facing filter** drops 89 events (USD 0.84 B) whose loss the earn user does not bear.
- Working sample $n = 1,075$, USD 9.45 B; 50% cross-source confirmed.

Why it matters

Prior work relies on a single feed, yet tail estimates are driven by the largest handful of events.

Tagged to the banking taxonomy

Every event carries a DeFi sub-sector and a **Basel Level-1 event type** — textual evidence first, feed classification as fallback.

Code	Basel Level-1	DeFi operational reading
IF	Internal Fraud	Rugpull, owner-drain, insider exit-scam, upgradeable-contract abuse.
EF	External Fraud	Contract bugs (reentrancy, access control), key compromise, DNS hijack.
CPBP	Clients, Products, Business Practices	Economic-design attacks: flashloan governance, oracle and price manipulation.
BDSF	Business Disruption, System Failures	Chain-level: MEV, sequencer outage, consensus issue, chain halt.
EDPM	Execution, Delivery, Process Mgmt	Team-side configuration and deployment errors.

EPWS and DPA have no on-chain analogue and are omitted. Seven sectors: Bridge, Lending, DEX, Yield, Stablecoin, Derivatives, Other.

Sector and event type are orthogonal

Aggregate gross loss, USD m

Sector	EF	IF	CPBP	EDPM	BDSF	Total
Bridge	3,084	127	6	8	12	3,237
Lending	704	568	928	14	8	2,222
Stablecoin	178	105	456	27	8	773
Yield	430	267	206	17	6	926
DEX	1,044	136	221	0	1	1,402
Derivatives	148	0	275	2	380	805
Other	35	4	48	0	0	87
Total	5,623	1,207	2,140	68	415	9,452

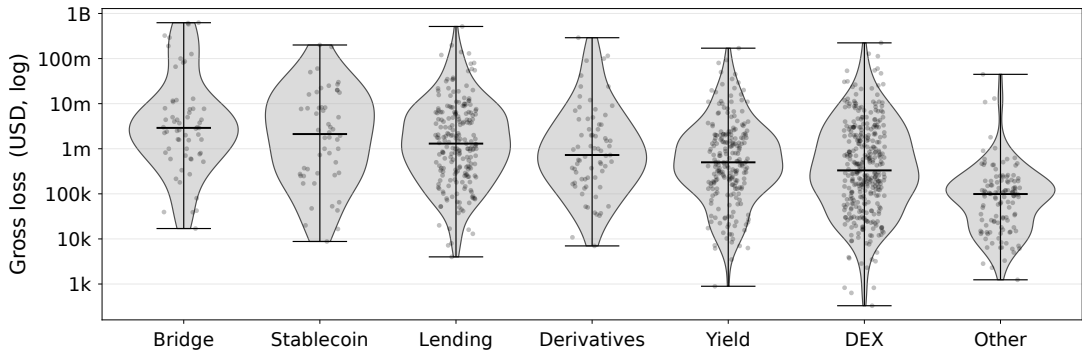
- Bridge and DEX are near-pure **EF** — code failing over pooled collateral. Bridge×EF alone is 33% of all losses.
- Lending and Stablecoin are **CPBP**-heavy — economic design, not code.
- Derivatives is the only sector with material **BDSF**.

Aggregate mean loss USD 8.79 m, median USD 0.50 m, max USD 624 m (Ronin).

Consequence for the model

Sector totals span two orders of magnitude and the loss mechanism differs by sector, so a single pooled fit would be meaningless. Everything is fitted **per sector**.

Loss distributions differ sharply across sectors



Violins encode density, bars the median. Bridge carries the highest median event loss (USD 2.90 m) from the pooled collateral bridges hold.

Method

The loss-distribution approach, per sector

Severity — peaks-over-threshold GPD on excesses $Y_i = X_i - u \mid X_i > u$:

$$g_{\xi, \beta}(y) = \frac{1}{\beta} \left(1 + \xi \frac{y}{\beta}\right)^{-(1/\xi+1)}$$

Threshold q^* by **plateau stability** ($\hat{\xi}$ stable to $\tau = 0.30$), subject to $n_u \geq 20$ and $n_u/n \geq 10\%$. The severity F_X is then piecewise: the **empirical body** below u^* , the **GPD tail** above it.

Frequency — monthly negative binomial $N_s \sim \text{NB}(\mu_s, \alpha_s)$; annual counts as twelve iid monthly draws.

Compound aggregation

$$S_{\text{year}} = \sum_{i=1}^N X_i, \quad N \sim \text{NB}(\hat{\lambda}, \hat{\alpha}), \quad X_i \stackrel{\text{iid}}{\sim} F_X$$

2×10^5 simulated years per sector, each event capped at the largest single-protocol exposure.

$\mathbb{E}[S]$ is the actuarial **pure premium**; $\text{VaR}_{99.9}$ is the **capital buffer** the Basel AMA would compute. Both expressed as a share of trailing-365d **TVL** — the on-chain analogue of assets on books.

From sector to protocol

No single protocol records enough events for its own fit — the same problem Basel solves for data-poor banks with a business-line severity allocated by an exposure indicator.

- Fit once per sector; treat its protocols as sharing that loss process **per unit of TVL** (credibility-theory shrinkage).
- A sector event strikes protocol p with probability $\text{TVL}_p/\text{TVL}_s$, so $\lambda_p = (\text{TVL}_p/\text{TVL}_s) \lambda_s$, same severity, capped at p 's own exposure.
- The tail does *not* thin linearly: with the single-loss approximation $\text{VaR}_{99.9}(S_p) \approx F_X^{-1}(1 - 0.001/\lambda_p)$ scaling as $\text{TVL}_p^{\hat{\xi}}$, $\hat{\xi} < 1$, the **VaR-to-TVL ratio rises as a venue shrinks**.

So do not scale the sector ratio

Simulating $\text{VaR}_{99.9}$ per venue rather than scaling the sector figure is what keeps the smaller venues from being understated.

The tail

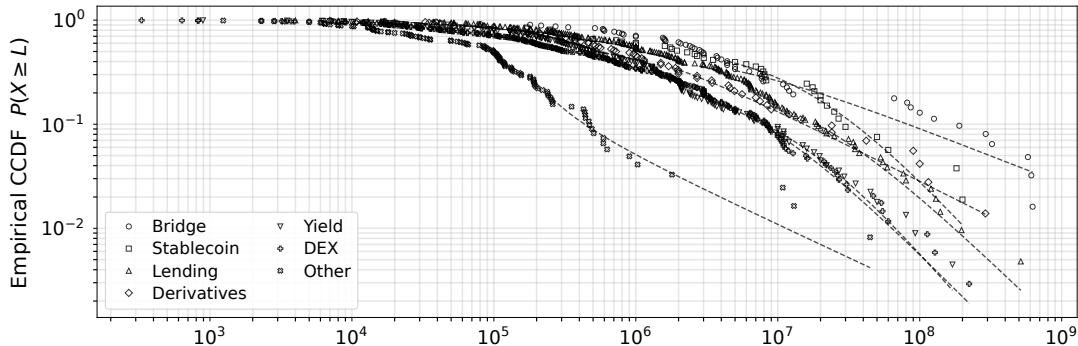
Severity: four sectors like a bank, three like cyber

Sector	n	q^*	n_u	$\hat{\xi}$ [95% CI]	$\hat{\beta}$ (USD m)	V	p
Bridge	61	0.65	20	+1.87 [+0.58, +3.18]	17.6	-1.37	0.17
Lending	207	0.90	21	+0.75 [-0.28, +1.50]	24.8	-1.32	0.19
Stablecoin	52	0.60	21	+0.65 [-0.39, +1.33]	13.2	-0.03	0.97
Derivatives	71	0.65	25	+1.45 [+0.42, +2.38]	3.7	-1.04	0.30
Yield	222	0.90	23	+0.61 [-0.22, +1.24]	11.5	-0.74	0.46
DEX	341	0.85	51	+0.72 [+0.18, +1.15]	7.1	-0.02	0.98
Other	121	0.80	24	+1.58 [+0.54, +2.63]	0.2	+0.57	0.57

Lending, Stablecoin, Yield, DEX cluster in $\hat{\xi} \in [0.61, 0.75]$ — **no heavier** than the Moscadelli banking band $[0.85, 1.39]$, and all finite-mean.

Bridge, Derivatives, Other reach $\hat{\xi} \approx 1.5$ – 1.9 — **cyber-loss level**, point estimates past the infinite-mean boundary. A Vuong test cannot separate GPD from lognormal in any sector ($|V| \leq 1.37$), so I keep the GPD.

The fitted tails



Log-log CCDF, loss in USD along the horizontal axis, fitted POT-GPD dashed. Where $\hat{\xi} > 1$ the fit turns up faster than the data at the top — what the exposure cap addresses.

Frequency: events cluster, so Poisson understates

Sector	n	$\hat{\mu}_s$	$\hat{\lambda}_s$	$\hat{\alpha}_s$	LR	p
Bridge	61	0.80	9.7	0.72	6.2	0.013
Lending	207	2.72	32.9	0.26	10.2	0.0014
Stablecoin	52	0.68	8.3	0.14	0.3	0.590
Derivatives	71	0.93	11.3	0.41	3.7	0.054
Yield	222	2.92	35.3	0.60	42.3	7.7×10^{-11}
DEX	341	4.49	54.2	0.31	38.5	5.4×10^{-10}
Other	121	1.59	19.2	0.89	30.4	3.6×10^{-8}

- Monthly counts are strongly over-dispersed: cross-sector dispersion index $D = 3.27$.
- A likelihood-ratio test rejects Poisson at $p \leq 0.02$ in **five of seven** sectors.
- Annual event rates run from 8.3/yr (Stablecoin) to 54.2/yr (DEX).

Under Poisson the annual aggregate — and its $\text{VaR}_{99.9}$ — would be understated.

The capital the tail implies

VaR_{99.9}, % of trailing-365d TVL

Sector	n	TVL (B)	VaR _{99.9}	VaR _{99.9} ^U	IQR
Bridge	61	49.1	32.5	67,326	[25.5, 40.8]
Lending	207	66.8	18.0	22.3	[3.2, 18.9]
Stablecoin	52	26.5 [†]	16.1	16.1	[3.1, 30.1]
Derivatives	71	65.9	22.8	638.0	[21.8, 27.1]
Yield	222	11.4	15.3	29.5	[5.5, 17.3]
DEX	341	14.5	14.1	46.2	[10.5, 15.9]
Other	121	18.2 [‡]	27.5	285.9	[4.8, 27.7]

[†] circulating supply removed [‡] residual TVL VaR^U: exposure cap

- Core sectors need **14–18% of TVL** — far above the operational-risk capital banks hold.
- Bridge, Derivatives, Other look larger (23–32%) but only because each event is **capped at exposure**.
- Uncapped, $\hat{\xi} > 1$ has no stable quantile: Bridge diverges to **673× TVL**.

The familiar pathology

The heaviest-tailed, most consequential sectors are exactly the ones the model cannot pin down unaided — the instability that retired Basel's internal-model AMA in 2017.

Sensitivity analysis

Sensitivity to the largest events. Dropping the single largest event moves $\hat{\xi}$ by 0.16–0.5, most sharply in Lending (0.75 $\rightarrow$ 0.35). Under any drop-top rule up to 0.5%, every sector keeps its **below / above infinite-mean** character.

Sensitivity to the cap. The cap binds: removing it lifts Lending 18 $\rightarrow$ 22%, DEX 14 $\rightarrow$ 46%, Yield 15 $\rightarrow$ 30%. Only Stablecoin is cap-insensitive.

Sensitivity to the confidence level. At 99.5% and 99% the Lending requirement falls to 7% and 5% of TVL, and buffered-venue coverage rises from 5% to 18% and 31%.

Conclusions survive

Even at 99%, the buffers cover about a third of a requirement that is still an order of magnitude above bank operational-risk capital.

Who bears it

The protocol's response: capital buffers

Top-10 Lending venues by gross supplied TVL (DefiLlama, June 2026): **four disclose an operational-risk reserve, six do not.**

Protocol	TVL (B)	Buffer (B)	$\mathbb{E}[S]$ (B)	VaR _{99.9} (B)
Aave V3	11.85	0.36	0.08 (475%)	4.00 (9%)
SparkLend	3.41	0.08	0.02 (390%)	1.58 (5%)
Compound V3	1.04	0.04	0.01 (700%)	0.64 (6%)
Venus Core Pool	0.98	0.01	0.01 (187%)	0.61 (2%)

Parentheses give the buffer's coverage of each. Average coverage of the modeled capital buffer: $\sim 5\%$, a 95% shortfall, from 2% (Venus) to 9% (Aave V3).

The inadequacy is specific to the tail

All four buffers cover the pure premium at least once over (187%–700%). They are sized to an average year, not to a 1-in-1,000 one.

The depositor's response: the yield premium

Risk premium = trailing-30d TVL-weighted stablecoin supply APY minus the 3.70% 3-month T-bill.

Protocol	Total yield	Risk premium	$\mathbb{E}[S]$	VaR _{99.9}
<i>No buffer</i>				
Morpho Blue	4.60 %	+90 bps	62 bps (144%)	3,905 bps (+2%)
JustLend V1	3.42 %	-28 bps	60 bps (-47%)	4,785 bps (-1%)
Kamino Lend	2.91 %	-79 bps	55 bps (-143%)	6,169 bps (-1%)
Jupiter Lend	4.64 %	+94 bps	55 bps (172%)	6,605 bps (+1%)
Fluid Lending	6.27 %	+257 bps	52 bps (497%)	6,507 bps (+4%)
Euler V2	3.59 %	-11 bps	47 bps (-23%)	8,131 bps (0%)
<i>With buffer</i>				
Aave V3	3.36 %	-34 bps	64 bps (-53%)	3,378 bps (-1%)
SparkLend	2.84 %	-86 bps	60 bps (-143%)	4,634 bps (-2%)
Compound V3	2.85 %	-85 bps	55 bps (-155%)	6,101 bps (-1%)
Venus Core Pool	2.09 %	-161 bps	55 bps (-295%)	6,226 bps (-3%)

Unbuffered venues average **+54 bps**: roughly their 55-bps pure premium, but 1–4% of the tail-risk premium — and the buffers of the previous slide cover $\mathbb{E}[S]$ several times over while covering 5% of VaR_{99.9}. **Protocol and depositor alike size their response to the loss of an average year, not to the extreme one.**

Both margins together

Because buffer and premium substitute, an efficient market would trade one for the other — buffered venues paying less. **The data run in that direction.**

- Median premium: -86 bps with a buffer against $+39$ bps without — a **125-bps gap**. One-sided Mann–Whitney $p = 0.01$, and $p = 0.03$ extending to all Lending venues with net TVL $\geq$ USD 100 m ($n = 19$).
- Below that size the signal attenuates: posted yields are dominated by airdrops and incentives. The long-tail depositor is **chasing incentives, not pricing risk**.

Right direction, wrong magnitude

The whole premium cross-section spans 418 bps, while the tail asks for about 18% of TVL. The market prices operational risk — and still leaves the tail overwhelmingly unfunded on both margins.

What follows

Who bears the shortfall, and what to do

The shortfall is not evenly borne. A professional participant can assess the tail, negotiate bilateral terms through governance, and size or exit the position. A **retail depositor sees only the posted rate** and, lacking standardized disclosure, bears the unpriced tail disproportionately.

Market discipline presumes the depositor *can* price the risk. For the retail depositor that presumption fails twice over: the information is not disclosed, and pricing a heavy tail takes skills they do not have.

So: disclosure, not capital mandates. There is no clear path to imposing bank-style capital requirements on autonomous protocols. Three measures instead:

1. **Standardized loss-and-yield disclosure** — a Pillar 3 analogue, per sector, alongside the posted yield.
2. **Buffer-and-tail disclosure** — the buffer, the share of $\text{VaR}_{99.9}$ it covers, the residual the depositor bears.
3. **Periodic recalibration** — the tail is non-stationary.

The duty can attach to whoever fronts access: front-ends, wallets, aggregators, custodial “earn” products.

Limitations and what I would fix next

- **Incomplete data.** Publicly reported losses only; missing small and privately resolved incidents bias the body down, leave the tail intact.
- **Homogeneity.** One severity per sector, allocated by TVL share. A size covariate or a hierarchical model would refine it.
- **Recovery.** Figures are gross of funds returned, so the capital estimates are conservative.
- **Confounding.** Buffered venues are also the largest and oldest — a descriptive association, not an identified buffer effect.
- **Premium identification.** The yield spread is reduced-form; it also reflects utilization and rate governance. A controlled panel regression is the natural extension.
- **Stationarity.** 2020–2026 fitted as one regime, though frequency has fallen and severity has concentrated.

Conclusion

- DeFi earn products carry a measurable operational-risk tail: USD 9.45 B over 1,075 events, with four core sectors no heavier-tailed than banking and three past the infinite-mean boundary.
- Calibrated as a supervisor would, the Lending tail asks for $\sim 18\%$ of TVL in capital.
- Against that benchmark, the four disclosed buffers cover 5%, and unbuffered venues pay +54 bps — short of the tail by more than an order of magnitude.
- The two margins move together in the right direction, so the market does discriminate; it just does not price.

The recommendation

Not capital mandates on autonomous protocols, but **disclosure** — standardized loss, yield and buffer-coverage information, owed by the protocol or by whoever fronts access to it.

Paper, data and code: github.com/nbundi/pricing-the-defi-tail ↗

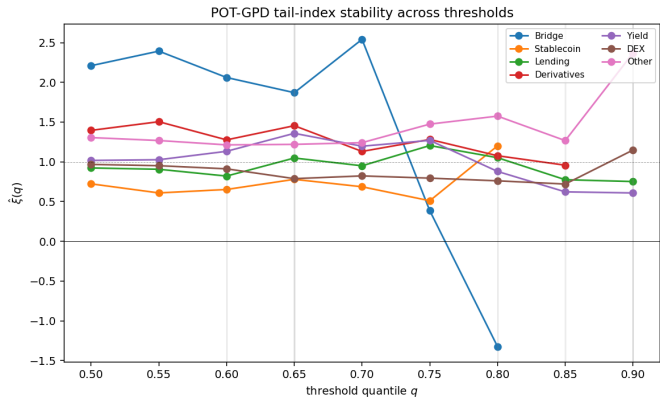
Questions and discussion.

Zurich University of
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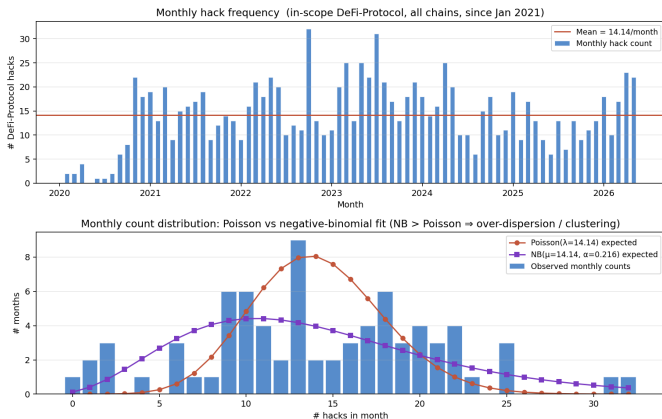
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Backup: threshold stability



$\hat{\xi}$ against the threshold quantile. q^* is the highest q at which $\hat{\xi}(q)$ is stable to within $\tau = 0.30$, subject to $n_u \geq 20$ and $n_u/n \geq 10\%$.

Backup: event frequency over time



Monthly event counts. Frequency has fallen from the early rugpull wave while severity has moved toward larger, concentrated targets — the non-stationarity the stationary fit averages over.

Backup: rolling intensity by sector

